

5.3 Context-Based Test Plan and Test Cases

The cases below are arranged by module and written around realistic user contexts. Each case identifies who is acting, the state of the system, the action performed, and the result that gives the function meaning. This format is more suitable for Ethernacare than a button-by-button checklist because safety behavior depends on time, ownership, verification, provider state, and the difference between local and server execution.

5.3.1 Test Cases

Table 5.3: Authentication and onboarding context cases

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
AUTH-01	A first-time user has an unused email address and accepts the current Terms.	Create an account with a valid name, email, and strong password; enter the 8-digit email code.	One auth identity and one public profile are created; setup continues only after confirmation.	Automated boundaries PASS; live email depends on configured SMTP.
AUTH-02	An account already exists for the same normalized email.	Attempt a second registration.	The system does not create a second account and shows a safe existing-account message.	Repository/auth handling PASS.
AUTH-03	A registered user forgot the password.	Request recovery, use the 8-digit recovery code or link, and set a compliant new password.	The recovery session is verified and the new password can be used to sign in.	Automated flow PASS; live email depends on SMTP.

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
AUTH-04	A user signed in through a configured OAuth provider for the first time.	Complete the provider redirect.	The auth UUID receives a public.users row and Supabase-backed modules load for the same user.	Bootstrap logic and diagnostics PASS.
AUTH-05	A signed-in user enabled biometric session unlock on a supported device.	Close and reopen the app; authenticate or cancel at the biometric prompt.	Successful local authentication unlocks the restored session; cancellation keeps the app locked.	Service tests PASS; physical sensor PENDING PHYSICAL.
AUTH-06	A first-time user has completed the required profile and primary-contact setup.	Step through or skip the guide, reopen it from Profile/Settings, then change reminder, sound, text-size, or reduced-motion settings.	The guide is offered once per account, remains replayable, and device settings persist and apply after navigation or restart.	Guide/settings widget and persistence tests PASS.

Table 5.4: Check-in, inactivity, and emergency context cases

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
SAFE-01	The user checked in less than one configured threshold ago.	Tap Oren again.	No duplicate heartbeat is created; status remains current and UI data refreshes.	RPC/fallback logic PASS.
SAFE-02	The threshold has elapsed since the latest heartbeat.	Tap Oren.	One new active check-in is inserted atomically and the rolling inactivity period restarts.	Automated threshold tests PASS.
SAFE-03	One full threshold window has been missed.	Allow the 15-minute server worker and the Android local monitor to evaluate inactivity.	The persisted status becomes not current; a local reminder is shown and no contact escalation occurs.	Logic and worker tests PASS; exact local timing PENDING PHYSICAL.
SAFE-04	Two full threshold windows have been missed and the user phone is verified.	Allow the next server worker evaluation.	One user-reminder SMS is queued/attempted, delivery state is recorded and no check-in is created.	Worker/orchestration PASS; carrier receipt PENDING PHYSICAL.

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
SAFE-05	Three full threshold windows have been missed and a verified primary contact exists.	Allow the third-stage server evaluation more than once.	One inactivity alert and one primary-contact SMS are queued/attempted; idempotency suppresses duplicate escalation.	Worker/idempotency PASS; carrier receipt PENDING PHYSICAL.
SAFE-06	The user presses manual SOS and grants location.	Confirm the SOS action.	The alert may include a Maps URL and may open the 999 dialer only through explicit user action.	Service/build PASS; GPS and SMS PENDING PHYSICAL.

Table 5.5: Trusted-contact and phone-verification context cases

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
CONTACT-01	A user has fewer than five contacts and enters a new international number.	Request and enter the 6-digit contact OTP, then save.	A recent contact_phone verification allows the owner-scoped insert.	Validation/service tests PASS; SMS PENDING PHYSICAL.
CONTACT-02	The number duplicates an existing contact or the user already has five contacts.	Attempt to save another contact.	The operation is rejected with a meaningful validation message.	Automated validation PASS.
CONTACT-03	A non-primary or unverified phone is entered on Legacy Check.	Submit the Legacy UID and phone.	No protected access is issued and the screen states that the phone is not the verified primary contact.	Legacy response/widget tests PASS.
CONTACT-04	Two contacts exist and ownership of the primary role changes.	Set the second contact as primary.	The atomic database operation leaves exactly one primary contact.	RPC/schema logic PASS.

Table 5.6: Oren, rewards, and admin context cases

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
OREN-01	Oren has 65 energy and no action occurs for three complete hours.	Reload Oren state after 3.5 hours.	Energy becomes 62, mood is recalculated, and partial hours are preserved for later decay.	Automated energy tests PASS.
OREN-02	Oren has 15 or less energy and the user owns a toy.	Press Play.	Oren refuses play, remains tired, and does not consume more energy.	Automated service tests PASS.
OREN-03	The user has enough tokens for an unowned toy.	Buy, select, sign out, sign in, and play with the toy.	Tokens decrease once; ownership and selection persist for that user on the device; the correct reaction is shown.	Service/cache tests PASS.
REWARD-01	A user meets a badge goal but has not opened the badge before.	Open the badge and select Collect Badge.	The badge is recorded in My Badge List once.	Service/migration tests PASS.
REWARD-02	A user meets a voucher goal.	Open Check Reward.	Reward details display a user-specific redeem code that remains stable for the earned voucher.	Service-boundary tests PASS.

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
ADMIN-01	An authorized reward admin is signed into the separate admin route.	Create or edit a reward, then observe another catalogue session.	Realtime updates the catalogue; manual refresh retrieves it if the subscription is delayed.	Admin tests PASS; live Realtime should be observed during acceptance.
ADMIN-02	A catalogue reward has no earned-user records, then another reward is already earned.	Delete each reward.	The unearned reward can be deleted; the earned reward is preserved or deactivated to protect history.	Admin deletion rules PASS.
OREN-04	Two Oren care or shop actions are initiated almost simultaneously.	Start overlapping Feed/Play or purchase/select operations.	Actions execute serially and both valid energy, token, ownership, and selection changes remain without a lost update.	Automated concurrency regression tests PASS.

Table 5.7: Legacy Planning and Legacy Checking context cases

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
LEGACY-01	The owner enabled Legacy Checking and the verified primary contact knows the UID and phone.	Verify with the 6-digit SMS code before day 90.	Funeral preferences are available; Legacy Notes and documents remain locked with clear remaining-day information.	Payload/widget tests PASS; SMS PENDING PHYSICAL.
LEGACY-02	The latest check-in reaches 90 days and no release is already active.	Allow the midnight cloud worker to run.	The owner receives a warning email and a 24-hour grace period; the primary contact does not yet receive protected access.	Worker/domain tests PASS; live email acceptance required.
LEGACY-03	The owner is inside the 24-hour grace period.	Check in or use the one-time Cancel Legacy Release link.	The pending release is revoked and protected content stays locked.	Cancellation function/tests PASS.
LEGACY-04	The owner does not cancel within 24 hours and all primary-contact eligibility rules still hold.	Run the next daily worker cycle.	The primary contact receives an email and the seven-day Notes/document window opens.	Worker logic PASS; live email acceptance required.

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
LEGACY-05	A seven-day window is open and the primary contact completes SMS verification.	View preferences, notes, and a secure document.	Only allowed owner data is returned; the document link is signed and short-lived.	Access filtering/signing tests PASS; end-to-end SMS PENDING PHYSICAL.
LEGACY-06	The seven-day window expires or the owner checks in.	Attempt another verification or document open.	Protected access is denied and stale release state is revoked.	Domain/access tests PASS.
LEGACY-07	The owner uploads a supported or unsupported file.	Try valid PDF/JPEG/PNG, oversized data, and mismatched extension/signature.	Only matching supported files at or below 10 MB are stored privately.	Document validation tests PASS.

Table 5.8: Non-functional context cases

ID	Context and precondition	Action / data	Expected outcome	Evidence / status
NFR-01	A mobile web user opens the release build at 390 x 844.	Load and scroll authentication and responsive content.	Viewport metadata is correct, content remains reachable, and no horizontal overflow occurs.	Release/mobile viewport PASS.
NFR-02	A different authenticated user modifies a client request with another user_id.	Attempt to read contacts, check-ins, rewards, notes, or documents.	RLS denies cross-user access.	Policy review and repository boundaries PASS; penetration testing remains future work.
NFR-03	Weather or AI provider is unavailable.	Open Home or AI Guidance.	Cached/fallback state is shown without blocking safety functions.	Automated fallback tests PASS.
NFR-04	An SMS provider stalls or a worker stops while a delivery is marked as processing.	Allow the provider call to exceed its limit or rerun the worker after the stale interval.	The request fails within 12 seconds, records a retryable state, can be reclaimed after five minutes, and rejects unverified recipients or unapproved messages.	Function logic/deployment PASS; live provider failure injection remains an acceptance test.

5.4 Chapter Summary and Evaluation

This chapter documented the current Ethernacare implementation. The application combines a layered Flutter client with Supabase Auth, PostgreSQL, Row Level Security, private Storage, Realtime and Edge Functions. A 15-minute cloud worker applies the rolling inactivity stages, while a daily cloud worker manages the 90-day Legacy sequence. The key rules are explicit: check-in eligibility follows the selected threshold; server-side SMS attempts occur at missed windows two and three; Oren energy decays hourly and care mutations are serialized; badges and vouchers have distinct redemption behaviour; biometric prompts are asynchronously coordinated; dashboard loads ignore stale overlapping responses; and protected Legacy content follows a 90-day, 24-hour owner-protection and seven-day contact-access sequence. The first-run guide and Settings module provide persistent onboarding and accessibility choices.

The 138-test suite and 30.64 percent line coverage provide useful regression evidence for deterministic logic and key interfaces. The deployed workers and versioned migrations provide an auditable backend implementation. However, automated tests and desktop builds cannot prove SMS receipt, external email delivery, Android operating-system scheduling, physical biometric sensors or GPS accuracy. Ethernacare is suitable for a final-year project and controlled demonstration, while production use requires the physical acceptance tests, provider monitoring and operational controls identified in this chapter.

Chapter 6

System Deployment

6 System Deployment

This chapter explains the controlled deployment of Ethernacare as a Flutter client with a managed Supabase backend. The project is not an industrial changeover from an earlier production system; deployment therefore focuses on repeatable cloud configuration, client builds, protected secrets, scheduled workers, rollback planning and user acceptance.

Deployment Architecture

The deployed production backend is Supabase. Authentication, PostgreSQL, Row Level Security, private Storage, Realtime, database functions and Edge Functions run in the configured Supabase project. The Python FastAPI worker stored in the repository is a readable reference implementation only and is not the production scheduler.

Client Deployment

The Flutter project can produce Android APK/app-bundle, Windows and web builds from the same Dart source. The public Supabase URL and anonymous key can be supplied at build time through `SUPABASE_URL` and `SUPABASE_ANON_KEY` definitions. Secret provider credentials are never compiled into the client. Release builds hide safety-test controls unless `ENABLE_SAFETY_TESTS` is explicitly enabled. Android deployment also requires notification, location and biometric permissions and the registered WorkManager entry point.

Cloud Functions and External Providers

Edge Functions implement phone OTP, emergency SMS, AI guidance, threshold processing, Legacy access, cancellation and Legacy inactivity processing. Supabase encrypted function secrets store the Twilio, transactional-email, AI and OTP-signing credentials. SMS calls use bounded provider timeouts and persisted retry state; interrupted processing can be reclaimed after a short stale-job window, and the emergency endpoint verifies both recipient eligibility and approved message categories. Twilio handles SMS delivery and Brevo/SMTP handles email; each provider must be configured, funded or verified according to its own account rules before real delivery can be accepted.

Database and Storage Deployment

Versioned SQL migrations create or alter tables, constraints, indexes, RPCs, RLS policies, Realtime publication settings, delivery outboxes and scheduled-job configuration. Migrations are applied in filename order and verified before the matching client release. The legacy-documents bucket is private, restricts file type and size, uses owner folders and returns short-lived signed URLs only through authorised flows.

Scheduled Worker Deployment

Supabase Cron invokes process-inactivity-thresholds every 15 minutes. The function recalculates missed windows for all eligible users, persists the latest stage, queues idempotent SMS work and retries eligible failures. A separate Cron job invokes process-legacy-inactivity at 12:00 a.m. Malaysia time each day to evaluate 90-day eligibility, owner warnings, cancellation grace, primary-contact release and seven-day expiry. Cron logs and Edge Function logs are checked after deployment.

System Backup and Risk Management

Source code and migrations are version-controlled. Database backups and provider exports should be retained according to the Supabase plan, while private files require a documented storage backup policy. Before a schema change, the developer reviews affected policies, functions and clients and prepares a backward-compatible or rollback migration. Secrets are stored only in encrypted project settings and must be rotated after accidental disclosure.

Major risks include provider outage, invalid SMS/email credentials, carrier filtering, delayed Android background work, revoked location permission, migration mismatch, stale processing jobs and duplicate alerts. The implementation reduces these risks with server scheduling, persisted outboxes, idempotency keys, bounded provider timeouts, failed-state retries, five-minute stale-job recovery, verified-recipient and message-category checks, friendly status messages, local fallbacks and audit logs. It cannot guarantee that an external message is delivered or read.

On-site Setup

A demonstration setup applies all migrations, deploys every Edge Function, configures required secrets, verifies both Cron jobs, enables authentication providers and redirect URLs, creates the private Storage bucket and builds the target Flutter package. Test accounts then complete email confirmation, phone verification, a primary contact, threshold configuration, biometric opt-in and Legacy consent. Production and testing phone numbers must be valid international numbers.

Training Procedure

Training begins with account creation and consent, followed by the skippable eight-step feature guide. It then demonstrates Oren check-in versus Feed/Play, the rolling threshold, reminder stages, contact verification, manual SOS, virtual rewards, profile controls and AI limitations. Users are shown how to replay the guide and use Settings for reminders, sounds, text size, reduced motion, biometrics and default restoration. A separate Legacy exercise covers the Legacy UID, funeral preferences, owner protection, the public contact flow and the fact that Notes/documents remain time-restricted. Administrators receive separate training for reward catalogue maintenance and provider/log monitoring.

Follow-up and Operational Monitoring

After release, the developer reviews Cron execution, Edge Function errors, SMS/email provider logs, failed outbox records, authentication delivery, Storage access and user-reported issues. Regression tests and static analysis are run before each release. Physical-device checks are repeated for notification permission, app resume, Doze/background behaviour, biometric cancellation/retry, GPS capture and carrier delivery.

Chapter Summary and Evaluation

EternaCare deployment separates a portable Flutter client from cloud-authoritative safety and Legacy workers. Versioned migrations, protected secrets, RLS, private Storage, scheduled functions and logs make the deployment repeatable and auditable. The approach resolves the main phone-background

limitation, but continued reliability still depends on provider configuration, operational monitoring and physical acceptance testing.